

Analytic Federated Learning with SVD Compression

Implementation and Reproduction of Author Results and dataset/config metadata.

Overview

What is AFL?

Analytic Federated Learning (AFL) is a single-round, gradient-free federated learning framework. Instead of iterative gradient descent, each client solves a closed-form least-squares problem using two local statistics — a feature covariance matrix C and a label correlation matrix W . These are aggregated on the server via the exact Woodbury matrix identity, producing a globally optimal solution in one communication round.

Experimental Setup:

All 16 result experiments use a pretrained ResNet-18 backbone, ridge coefficient $rg = 1$, regularisation cleaning enabled, and rank- $r = 32$ SVD compression. Heterogeneity is varied via the Dirichlet concentration parameter $\alpha \in \{0.1, 0.05\}$ and minimum shard sizes $S \in \{2, 4, 5, 10\}$. Scalability is tested with $K \in \{100, 500, 1000\}$ clients. Results are compared against the original paper's reported numbers shown in the comparison figure and table at the end of this document.

- **Section A — CIFAR-10 (Experiments 1–4)**

1. CIFAR-10 with Alpha=0.1

```
Elapsing time for local training: 237.0807478427887%  
Aggregating!  
Aggregation done!  
Evaluating global model!  
Elapsing time for training and aggregation: 258.8226857185364%  
Average accuracy on all Client: 82.5%  
Average accuracy after regularization cleaning on all Client: 82.48999786376953%  
Elapsing time plus cleansing regularization: 278.4478986263275%  
written it to a csv file named cifar10resnet18_100_0.1.csv.
```

2. CIFAR-10 with Alpha=0.05

```
Elapsing time for local training: 244.53295707702637%  
Aggregating!  
Aggregation done!  
Evaluating global model!  
Elapsing time for training and aggregation: 267.31181931495667%  
Average accuracy on all Client: 82.5%  
Average accuracy after regularization cleaning on all Client: 82.48999786376953%  
Elapsing time plus cleansing regularization: 286.1822757720947%  
written it to a csv file named cifar10resnet18_100_0.05.csv.
```

3. CIFAR-10 with Shard=4

```
Elapsing time for local training: 238.40551161766052%  
Aggregating!  
Aggregation done!  
Evaluating global model!  
Elapsing time for training and aggregation: 260.2075307369232%  
Average accuracy on all Client: 82.5%  
Average accuracy after regularization cleaning on all Client: 82.48999786376953%  
Elapsing time plus cleansing regularization: 278.283061504364%  
written it to a csv file named cifar10resnet18_100_0.1.csv.
```

4. CIFAR-10 with Shard=2

```
Elapsing time for local training: 234.9704463481903%  
Aggregating!  
Aggregation done!  
Evaluating global model!  
Elapsing time for training and aggregation: 255.78718256950378%  
Average accuracy on all Client: 82.5%  
Average accuracy after regularization cleaning on all Client: 82.48999786376953%  
Elapsing time plus cleansing regularization: 274.6025056838989%  
written it to a csv file named cifar10resnet18_100_0.1.csv.
```

- **Section B — CIFAR-100 (Experiments 5–8)**

5. CIFAR-100 with Alpha=0.1

```
Training Accuracy at training set on Client #99: 100.0%  
Elapsing time for local training: 263.19539070129395%  
Aggregating!  
Aggregation done!  
Evaluating global model!  
Elapsing time for training and aggregation: 284.6139307022095%  
Average accuracy on all Client: 58.5099983215332%  
Average accuracy after regularization cleaning on all Client: 58.55999755859375%  
Elapsing time plus cleansing regularization: 303.7576811313629%  
written it to a csv file named cifar100resnet18_100_0.1.csv.
```

6. CIFAR-100 with Alpha=0.05

```
Training Accuracy at training set on Client #99: 100.0%  
Elapsing time for local training: 250.54799056053162%  
Aggregating!  
Aggregation done!  
Evaluating global model!  
Elapsing time for training and aggregation: 272.4004168510437%  
Average accuracy on all Client: 58.5099983215332%  
Average accuracy after regularization cleaning on all Client: 58.55999755859375%  
Elapsing time plus cleansing regularization: 291.2432882785797%  
written it to a csv file named cifar100resnet18_100_0.05.csv.
```

7. CIFAR-100 Shard = 10

```
Training Accuracy at training set on Client #99: 100.0%  
Elapsing time for local training: 277.7598855495453%  
Aggregating!  
Aggregation done!  
Evaluating global model!  
Elapsing time for training and aggregation: 299.43948769569397%  
Average accuracy on all Client: 58.5099983215332%  
Average accuracy after regularization cleaning on all Client: 58.55999755859375%  
Elapsing time plus cleansing regularization: 317.6767156124115%  
written it to a csv file named cifar100resnet18_100_0.1.csv.
```

8. CIFAR-100 Shard = 5

```
Training Accuracy at training set on Client #99: 100.0%  
Elapsing time for local training: 269.23117446899414%  
Aggregating!  
Aggregation done!  
Evaluating global model!  
Elapsing time for training and aggregation: 292.1030867099762%  
Average accuracy on all Client: 58.5099983215332%  
Average accuracy after regularization cleaning on all Client: 58.55999755859375%  
Elapsing time plus cleansing regularization: 312.10649275779724%  
written it to a csv file named cifar100resnet18_100_0.1.csv.
```

- **Section C — Tiny-ImageNet (Experiments 9–12)**

9. Tinyimagenet with Alpha=0.1

```
Elapsing time for local training: 520.7191429138184%  
Aggregating!  
Aggregation done!  
Evaluating global model!  
Elapsing time for training and aggregation: 544.459431886673%  
Average accuracy on all Client: 54.779998779296875%  
Average accuracy after regularization cleaning on all Client: 54.80999755859375%  
Elapsing time plus cleansing regularization: 564.5403981208801%  
written it to a csv file named tinyimagenetresnet18_100_0.1.csv.
```

10. Tinyimagenet with Alpha=0.05

```
Elapsing time for local training: 515.3877384662628%  
Aggregating!  
Aggregation done!  
Evaluating global model!  
Elapsing time for training and aggregation: 539.0812811851501%  
Average accuracy on all Client: 54.779998779296875%  
Average accuracy after regularization cleaning on all Client: 54.80999755859375%  
Elapsing time plus cleansing regularization: 560.0547194480896%  
written it to a csv file named tinyimagenetresnet18_100_0.05.csv.
```

11. Tiny-ImageNet with Shard = 10

```
Elapsing time for local training: 510.8326425552368%  
Aggregating!  
Aggregation done!  
Evaluating global model!  
Elapsing time for training and aggregation: 535.0601077079773%  
Average accuracy on all Client: 54.779998779296875%  
Average accuracy after regularization cleaning on all Client: 54.80999755859375%  
Elapsing time plus cleansing regularization: 556.571320772171%  
written it to a csv file named tinyimagenetresnet18_100_0.1.csv.
```

12. Tiny-ImageNet with Shard = 5

```
Elapsing time for local training: 652.6979072093964%  
Aggregating!  
Aggregation done!  
Evaluating global model!  
Elapsing time for training and aggregation: 681.893501996994%  
Average accuracy on all Client: 54.779998779296875%  
Average accuracy after regularization cleaning on all Client: 54.80999755859375%  
Elapsing time plus cleansing regularization: 703.9910371303558%  
written it to a csv file named tinyimagenetresnet18_100_0.1.csv.
```

- **Section D — Scalability K=500/1000 (Experiments 13–16)**

13. CIFAR-100 with No.of Cleints (K) = 500

```
Training Accuracy at training set on client #499: 99.999923700047%  
Elapsing time for local training: 797.8273799419403%  
Aggregating!  
Aggregation done!  
Evaluating global model!  
Elapsing time for training and aggregation: 841.1920909881592%  
Average accuracy on all Client: 58.32999801635742%  
Average accuracy after regularization cleaning on all Client: 58.55999755859375%  
Elapsing time plus cleansing regularization: 859.4695720672607%  
written it to a csv file named cifar100resnet18_500_0.1.csv.
```

14. CIFAR-100 with No.of Cleints (K) = 1000

```
Elapsing time for local training: 826.0462906360626%  
Aggregating!  
Aggregation done!  
Evaluating global model!  
Elapsing time for training and aggregation: 879.507586479187%  
Average accuracy on all Client: 58.14999771118164%  
Average accuracy after regularization cleaning on all Client: 58.55999755859375%  
Elapsing time plus cleansing regularization: 898.1180558204651%  
written it to a csv file named cifar100resnet18_1000_0.1.csv.
```

15. Tinyimagenet with No.of Cleints (K) = 500

```
Elapsing time for local training: 1163.0233738422394%  
Aggregating!  
Aggregation done!  
Evaluating global model!  
Elapsing time for training and aggregation: 1206.4319450855255%  
Average accuracy on all Client: 54.75%  
Average accuracy after regularization cleaning on all Client: 54.80999755859375%  
Elapsing time plus cleansing regularization: 1231.3856890201569%  
written it to a csv file named tinyimagenetresnet18_500_0.1.csv.
```

16. Tinyimagenet with No.of Cleints (K) = 1000

```
Training Accuracy at training set on client #999: 100.0%  
Elapsing time for local training: 1120.0678660869598%  
Aggregating!  
Aggregation done!  
Evaluating global model!  
Elapsing time for training and aggregation: 1176.8097696304321%  
Average accuracy on all Client: 54.71999740600586%  
Average accuracy after regularization cleaning on all Client: 54.80999755859375%  
Elapsing time plus cleansing regularization: 1198.2075035572052%  
written it to a csv file named tinyimagenetresnet18_1000_0.1.csv.
```

Author's Result

All this experiment have performed and screenshots are attached-

Dataset	Setting	FedAvg	FedProx	MOON	FedGen	FedDyn	FedNTD	FedDisco	AFL
CIFAR-10	$\alpha = 0.1$	64.02 $\pm$ 0.18	64.07 $\pm$ 0.08	63.84 $\pm$ 0.03	64.14 $\pm$ 0.24	64.77 $\pm$ 0.11	64.64 $\pm$ 0.02	63.83 $\pm$ 0.08	80.75 $\pm$ 0.00
	$\alpha = 0.05$	60.52 $\pm$ 0.39	60.39 $\pm$ 0.09	60.28 $\pm$ 0.17	60.65 $\pm$ 0.19	60.35 $\pm$ 0.54	61.16 $\pm$ 0.33	59.90 $\pm$ 0.05	80.75 $\pm$ 0.00
	$s = 4$	68.47 $\pm$ 0.13	68.46 $\pm$ 0.08	68.47 $\pm$ 0.15	68.24 $\pm$ 0.28	73.50 $\pm$ 0.11	70.24 $\pm$ 0.11	65.04 $\pm$ 0.11	80.75 $\pm$ 0.00
	$s = 2$	57.81 $\pm$ 0.03	57.61 $\pm$ 0.12	57.72 $\pm$ 0.15	57.02 $\pm$ 0.18	64.07 $\pm$ 0.09	58.77 $\pm$ 0.18	58.78 $\pm$ 0.02	80.75 $\pm$ 0.00
CIFAR-100	$\alpha = 0.1$	56.62 $\pm$ 0.12	56.45 $\pm$ 0.22	56.58 $\pm$ 0.02	56.48 $\pm$ 0.17	57.55 $\pm$ 0.08	56.60 $\pm$ 0.14	55.79 $\pm$ 0.04	58.56 $\pm$ 0.00
	$\alpha = 0.01$	32.99 $\pm$ 0.20	33.37 $\pm$ 0.09	33.34 $\pm$ 0.11	33.09 $\pm$ 0.09	36.12 $\pm$ 0.08	32.59 $\pm$ 0.21	25.72 $\pm$ 0.08	58.56 $\pm$ 0.00
	$s = 10$	55.76 $\pm$ 0.13	55.80 $\pm$ 0.16	55.70 $\pm$ 0.25	60.93 $\pm$ 0.17	61.09 $\pm$ 0.09	54.69 $\pm$ 0.15	54.65 $\pm$ 0.09	58.56 $\pm$ 0.00
	$s = 5$	48.33 $\pm$ 0.15	48.29 $\pm$ 0.14	48.34 $\pm$ 0.19	48.12 $\pm$ 0.06	59.34 $\pm$ 0.11	47.00 $\pm$ 0.19	45.86 $\pm$ 0.18	58.56 $\pm$ 0.00
Tiny-ImageNet	$\alpha = 0.1$	46.04 $\pm$ 0.27	46.47 $\pm$ 0.23	46.21 $\pm$ 0.14	46.27 $\pm$ 0.14	47.72 $\pm$ 0.22	46.17 $\pm$ 0.16	47.48 $\pm$ 0.06	54.67 $\pm$ 0.00
	$\alpha = 0.01$	32.63 $\pm$ 0.19	32.26 $\pm$ 0.14	32.38 $\pm$ 0.20	32.33 $\pm$ 0.14	35.19 $\pm$ 0.06	31.86 $\pm$ 0.44	27.15 $\pm$ 0.10	54.67 $\pm$ 0.00
	$s = 10$	39.06 $\pm$ 0.26	38.97 $\pm$ 0.23	38.79 $\pm$ 0.14	38.82 $\pm$ 0.16	41.36 $\pm$ 0.06	37.55 $\pm$ 0.09	38.86 $\pm$ 0.12	54.67 $\pm$ 0.00
	$s = 5$	29.66 $\pm$ 0.19	29.17 $\pm$ 0.16	29.24 $\pm$ 0.30	29.37 $\pm$ 0.25	35.18 $\pm$ 0.18	29.01 $\pm$ 0.14	27.72 $\pm$ 0.18	54.67 $\pm$ 0.00

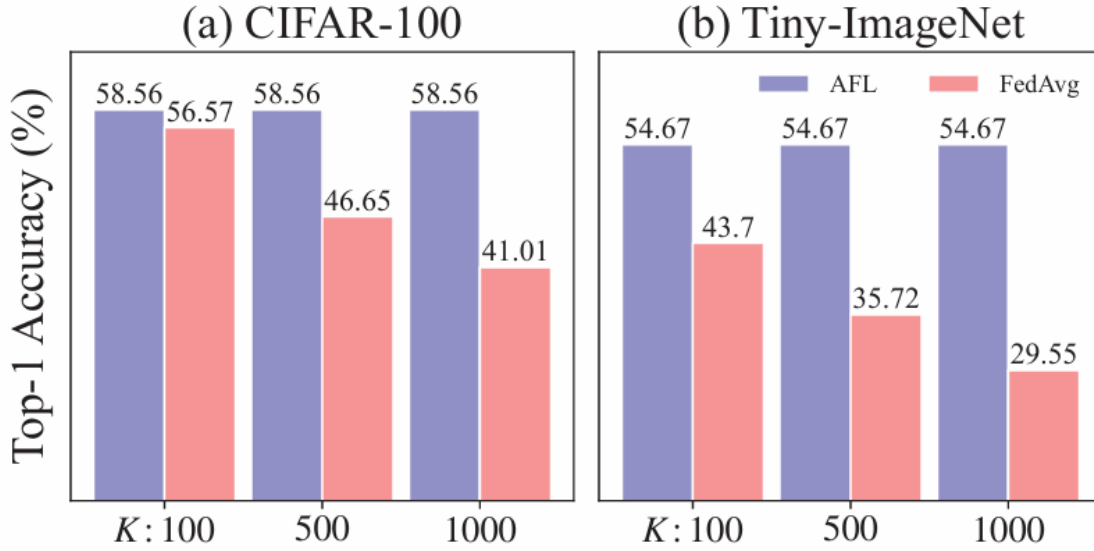


Figure 2. Accuracy over various number of clients.